Answer Key

QPCODE: 25900293

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) – MG3DSECHEA03 – PRODUCTION OPERATIONS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Part A

Short Answer Questions Answer any **5** questions

Each questions carries **4** marks

1. Differentiate between thermal cracking and catalytic cracking. [U] / [CO1]

   Thermal Cracking It uses high temperature (450–750°C) and high pressure (up to 70 atm) to break large hydrocarbon molecules. The products have lower octane number and poorer quality. Less efficient and requires more energy. Sulphur content of thermal cracked gasoline is higher.

   Catalytic Cracking: It uses a catalyst (like zeolite or silica–alumina) to promote cracking at lower temperature and low pressure. The process is highly selective, producing more gasoline and less gas or coke. The products have higher octane number, making them better for fuels. More efficient and economical due to catalyst use. Sulphur content of catalytically cracked gasoline is lower than thermally cracked gasoline. **(Any two points from each cracking-4 Marks)**

2. Name any four factors that influence hydrocracking efficiency. [K] / [CO2]
   - Feedstock characteristics such as sulfur, nitrogen, metal content, aromaticity, and hydrogen-to-carbon ratio play a major role.
   - Reaction conditions including temperature, pressure, and hydrogen-to-hydrocarbon ratio affect the rate of cracking and hydrogenation.
   - Catalyst type and activity are crucial, as hydrocracking uses bifunctional catalysts with both acidic and hydrogenation functions, and their activity determines selectivity and yield.
   - By carefully controlling these factors, refiners can optimize product yield, quality, and catalyst life in hydrocracking units. **(Any Four points-4 Marks)**

3. List any four advantages of catalytic cracking over thermal cracking. [K] / [CO2]
   - Catalytic cracking occurs at lower temperature and pressure compared to thermal cracking, saving energy and cost.
   - Catalytically cracked gasoline consists of isoparaffins and aromatics which are chemically stable than gasoline obtained from thermal cracking.
   - Catalytically cracked gasoline has high octane number. Produces fewer unwanted gases

and coke; product distribution can be controlled more easily.

- Sulphur content of catalytically cracked gasoline is lower than thermally cracked gasoline. Catalysts can be regenerated and reused, making the process more economical and continuous. **(Any Four points-4 Marks)**

4. Discuss delayed coking and its advantages. [U] / [CO3]
   Delayed coking is a semi-batch thermal cracking process where preheated heavy oils are sent to a coke drum, where cracking and coke formation take place. In this process, heavy residual oil is cracked in coke drums to form solid coke and vapors. Solid coke remains in the coke drum, once full, the drum is cooled and the solid coke is cut out. Stream and volatiles escape from the coke drum is again fed to the fractionator. These vapors give heat to the incoming fed and it contain valuable light hydrocarbons. Delayed coking controls the coke formation inside large coke drums, ensures better heat management, gives higher yield of liquid products, and allows continuous operation with periodic coke removal.

   (2 marks each for explanation and any 2 marks for advantages)

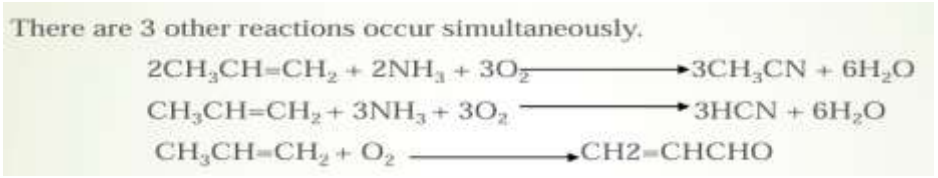
5. What are clathrate hydrates? Mention two problems they cause in petroleum industries. [U] / [CO4]
   Hydrates can be defined as solid crystalline compounds formed by hydrocarbon or metals and H_2O under reduced temperature and pressure. Hydrates have a system of cavities. They are also termed as clathrates. A clathrate hydrate is a special type of gas hydrate in which lattice of water molecules enclose molecules of a trapped gas. **Problems:**

- Clathrate hydrates can form solid ice-like plugs inside pipelines, restricting or completely blocking the flow of oil and gas.
- The formation of hydrates disrupts steady flow in subsea and production lines, leading to pressure drops and operational inefficiency.
- Hydrate blockages often force shutdowns of production facilities, leading to significant downtime and economic losses. **(2 marks each for definition and any two problems)**

6. What are acidic and basic oxides? Give one examples [K] / [CO4]
   Acidic oxides: They are obtained by oxidation of nonmetals. They are easily volatile. The H_2O solution is acidic and they are covalent bond compounds. eg:- CO_2, SO_3, P_4O_{10} etc.
   Basic oxides: They obtained by oxidising metals. They are non-volatile because it has high boiling point. The alkaline. They are ionic compounds. eg:- Na_2O, MgO, CaO etc.

   (2 marks each for definition and one example)

7. Write the balanced chemical equations for acrylonitrile production by ammoxidation and identify the major side reaction. [A] / [CO4]
   $2CH_3CH{=}CH_2 + 2NH_3 + 3O_2 \longrightarrow 2CH_2{=}CH\text{-}CN + 6H_2O$
   This is a one-step reaction between propylene, ammonia and oxygen



(1 mark each for each equation)

Part B

Short Essay Questions

Answer any **5** questions.

Each question carries **7** marks.

8. Discuss the Industrial importance of thermal cracking in the petrochemical industry. Give examples of products obtained and their uses. [U] / [CO1]

Industrial applications of thermal cracking include the production of gasoline, olefins, diesel, LPG, visbreaking of residues, and carbon black manufacture. It plays a major role in both fuel production and petrochemical industries.

- Thermal cracking of heavy oils (like gas oil, vacuum residues) produces high-octane gasoline, which is used as motor fuel. Light hydrocarbons (C_3–C_4 gases like propane, butane) are formed as by-products of thermal cracking. Steam cracking, is the main source of ethylene, propylene, and butadiene. These are essential building blocks for plastics (polyethylene, polypropylene, synthetic rubber, PVC). Helps improve fuel oil quality and reduce pour point. Some thermal cracking processes yield middle distillates like diesel, kerosene, and jet fuel and LPG. **(3 marks for Ist part and 2 marks for examples and uses)**

9. Discuss the limitations of fixed bed and moving bed processes and how fluidized bed cracking overcomes them. [U] / [CO2]

Fixed bed catalyst: Continuous operation is not possible (requires shutdown for regeneration). Limited catalyst utilization. Not suitable for continuous large-scale operations.

Moving bed catalyst: Complex design and higher cost. Catalyst attrition (wear and tear) due to movement. Possibility of catalyst loss or contamination during movement

Fluidized Bed Cracking Overcomes these Limitations:

Provides continuous catalyst regeneration, maintaining high activity. Ensures better mixing and contact between feed and catalyst, improving cracking efficiency. Allows uniform temperature control and smoother operation, leading to higher yield and process stability. Catalyst regeneration without shutdown. **(2 marks each for the limitations of fixed bed and moving bed and 3 marks for advantages of fluidized bed cracking- ANY 2 points for each process)**

10. Discuss the desirable properties of commercial hydrocracking catalysts. [U] / [CO2]

Catalyst type and activity are crucial, as hydrocracking uses bifunctional catalysts with both acidic and hydrogenation functions, and their activity determines selectivity and yield. The catalyst should efficiently promote both cracking and hydrogenation reactions under moderate temperature and pressure. It should selectively produce desired products such as middle distillates (diesel, kerosene) with minimal unwanted by-products. The catalyst must withstand high temperatures, pressures, and mechanical stress without breaking down. It should resist deactivation by impurities such as sulfur, nitrogen, and metals present in the feedstock. **(Any FIVE Points Give 7 Marks)**

11. Explain the manufacture of petroleum coke by thermal cracking. [U] / [CO3]

Thermal cracking is a non-catalytic process in which heavy petroleum fractions (like vacuum residue) are broken into lighter hydrocarbons and solid coke by applying high temperature and pressure in the absence of oxygen. This process is done in Dubs cracking unit. These units have 2 sections and it is consisting of evaporators and coke drums. The feed is initially preheated in a furnace to a temperature of around 450–500°C. The heated feed then enters a coke drum or reaction chamber, where thermal cracking takes place. Due to the high temperature, the large hydrocarbon molecules break down into lighter fractions such as gases, oils, and solid coke. Importantly, the process is carried out in the absence of air to prevent combustion. The vapor products from the cracking reaction are directed to a fractionating tower, where they are separated into different components such as LPG, gasoline, diesel, and gas oil, while the solid coke settles at the bottom of the drum. Once the coke drum becomes full, it is cooled, and the accumulated coke is removed using high-pressure water jets. The

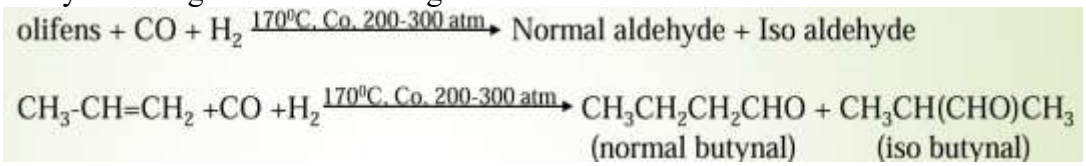
removed coke is then crushed, screened, and stored for further use or disposal. **(Proper explanation of the method 7 Marks)**

12. Describe the modified hydroformylation process using rhodium catalyst. How does it improve yield compared to cobalt catalyst process? [A] / [CO4]

In the modified hydroformylation (oxo) process**,** rhodium-based catalysts (hydrocarbonyl tris rhodium) are used instead of cobalt catalysts. The reaction involves the addition of carbon monoxide and hydrogen to an olefin to produce aldehydes**.** The process operates at lower temperature and lower pressure compared to the cobalt-catalyzed process. There action is 100 times faster than the conventional process. The ratio of normal to iso aldehyde is16:1 more over catalyst can be recovered completely. The rhodium-catalyzed modified hydroformylation process offers higher yield, better selectivity, and lower operating costs compared to the older cobalt-catalyzed process. **(Explanation of modified method 5 marks & Proper explanation of the 2nd part 2 marks)**

13. Explain the hydrogenation of oxo-aldehydes. Write the reaction and discuss its industrial importance. [A] / [CO4]

This is a process by which an olefine reacts with Hydrogen and carbon monoxide to produce an aldehyde having a carbon atom higher than the olefine used.



The aldehyde obtained is catalytically hydrogenated over nickel catalyst to produce alcohol.



Produces fatty alcohols, which are key raw materials for detergents, surfactants, and plasticizers. Converts aldehydes (which are less stable and reactive) into more stable and useful alcohols. **(Explanation & Equation 5 marks & Application 2 marks)**

14. Explain the role of syngas and aromatic compounds in the petrochemical industry. [U] / [CO1]

Syngas provides basic building blocks for many chemical syntheses.

Syngas is a mixture of carbon monoxide (CO) and hydrogen (H_2) and is produced by gasifying carbon-based feedstocks (e.g., coal, natural gas, biomass).

Products Derived from Syngas:

Methanol (CH_3OH): Used in producing chemicals like formaldehyde, acetic acid, and plastics.

Ammonia (NH_3): used to produce fertilizers, explosives, and plastics.

Aromatics: Aromatic hydrocarbons such as benzene, toluene, and xylenes are produced primarily through the refining of naphtha or by catalytic reforming of petroleum.

Products Derived from Aromatics:

Benzene (C_6H_6): Used in the production of styrene (for polystyrene), phenol, cyclohexane, and nylon. Toluene ($C_6H_5CH_3$): Solvent, and a precursor to benzene, xylene, and toluene diisocyanate (used in polyurethane production). Xylene ($C_6H_4(CH_3)_2$): Used in the production of polyester fibers, solvents, and paints. **(3 points each from syn gas and aromatics 7 marks)**

[7x5 = 35]

Part C

Essay Questions Answer any **1** question.
Each question carries **15** marks.

15. Identify the role of catalytic cracking in modern refinery operations. [K] / [CO2]

Catalytic cracking operates on the principle of breaking large hydrocarbon molecules into

smaller ones using a suitable catalyst, typically a zeolite-based material. The process occurs at relatively lower temperatures and pressures compared to thermal cracking, due to the presence of the catalyst which promotes efficient bond breaking and rearrangement of hydrocarbon chains. Catalytic cracking proceeds via ionic mechanism involving formation of carbocation. There are 3 types of catalytic cracking processes- fixed bed cracking, moving bed process, fluidised bed process. The most common industrial unit used today is the Fluidized Catalytic Cracking, which allows continuous catalyst regeneration and high conversion rates. Advantages: Catalytically cracked gasoline consists of isoparaffins and aromatics which are chemically stable than gasoline obtained from thermal cracking. Catalytically cracked gasoline has high octane number. Sulphur content of catalytically cracked gasoline is lower than thermally cracked gasoline. Catalytic cracking gives rise to lesser number of residues.

(Catalytic cracking, its advantages and applications give 5 marks each)

16. Explain the principle, types, and chemical reaction involved in reforming. [U] / [CO3]

Reforming is a petroleum refining process that converts low-octane straight-chain hydrocarbons (paraffins and naphthenes) into high-octane aromatic hydrocarbons and branched-chain alkanes.

There are two types of reforming- a) Thermal reforming b) Catalytic reforming

Reforming works based on four key reactions:

1. Dehydrogenation of naphthenes to aromatics
2. Isomerization of Paraffins
3. Dehydrocyclization of Paraffins
4. Hydrocracking (minor reaction)
5. Dealkylation

(Principle: 2 marks, Types: 5 marks, Reactions: 7 marks)

[15x1 = 15]

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